A.5 Inference Settings for Models

Model	Temperature
GPT-4o (August 2024)	0.0
Llama 3.1 405B Instruct	0.2
Mistral Large (November 2024)	0.2
Claude 3.5 Sonnet (October 2024)	0.2
Gemini 1.5 Pro (August 27, 2024)	1.0
o1-preview	1.0
Llama-3.2-3B-Instruct	0.6
Llama-3.3-70B-Instruct	0.6
Qwen2-72B-Instruct	0.7
Qwen2.5-14B-Instruct	0.7
Qwen2.5-72B-Instruct	0.7
Mixtral-8x7B-Instruct-v0.1	0.7
Mixtral-8x22B-Instruct-v0.1	0.7

Table 6: Inference temperatures for LLMs evaluated on `MultiChallenge`.

A.6 Model Performance over time

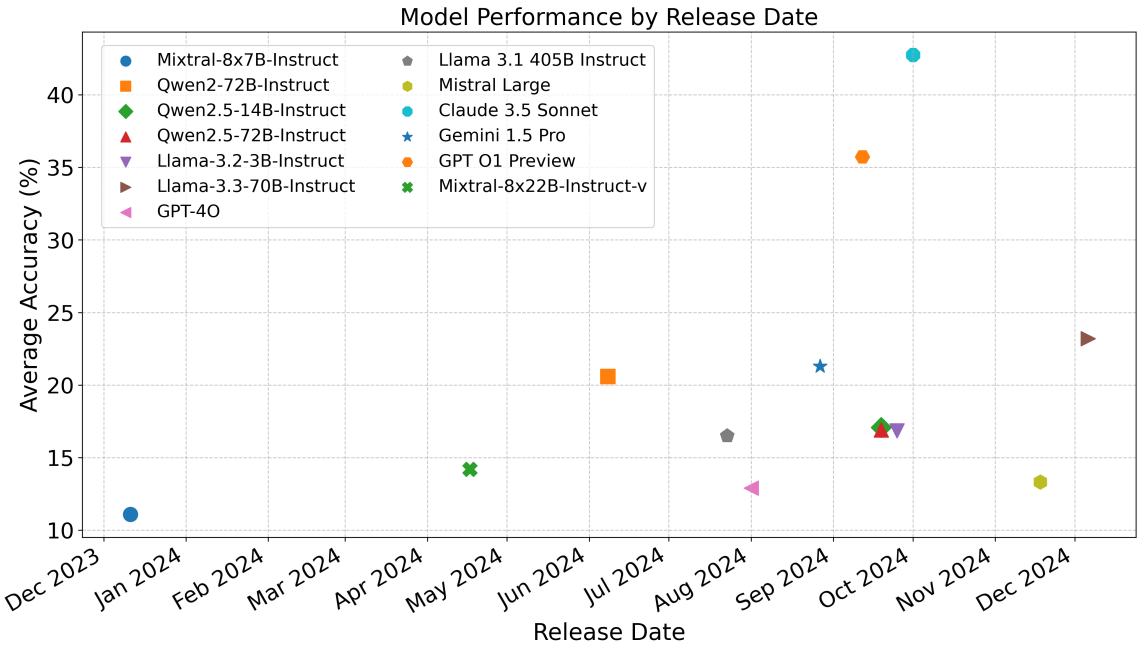

Figure 8: Model performance over time, based on release dates.

A.7 Conversation Blueprint Examples

Blueprint for Testing the Model's Recollection Ability

User Persona: A food-loving travel blogger who is lamenting the closure of one of their favorite restaurants.

Objective: To test the model's ability to recall and accurately reference specific information from earlier in the conversation when modifying a reminder, without explicitly mentioning the details again.

Conversation Plan:

Turn 1 (User):

- **Action:** Express sadness over the recent closure of their favorite restaurant.
- **Details to Include:**
  - Name of the restaurant: **"La Petite Table"**.
  - Location: **Marseille, France**.
  - Signature dish: **Bouillabaisse with a modern twist**.
  - Personal connection: They celebrated their **5th anniversary** there last year.
  - Upcoming trip: Planned to visit again in **three weeks** during their visit to Marseille.

Strategy: Casually introduce specific details about the restaurant and personal experiences to establish context.

Turn 2 (Assistant):

- Responds with empathy, acknowledging the user's feelings, and perhaps asks about what made the restaurant special to them.

Turn 3 (User):

- **Action:** Share more about what made "La Petite Table" special, perhaps mentioning the **chef's unique approach** and the **ambiance**.
- **Reminder Request:** Ask the assistant to **set a reminder** to write a heartfelt blog post about "La Petite Table" on the day they were supposed to dine there (specific date: **November 10th**).

Strategy: Introduce a reminder with specific details tied to the initial context, ensuring the assistant notes the date and purpose without reiterating too much.

… (Turns 4-7 omitted for brevity)

Turn 8 (Assistant):

- Responds positively, perhaps offers to set a new reminder or provide more information about "Le Nouveau Bistro".

Turn 9 (User):

- **Action:** Request to **modify the original reminder**.
- **Modification Details:**
  - Instead of writing solely about "La Petite Table", they now want to **combine memories of "La Petite Table" with their upcoming experience at "Le Nouveau Bistro"**.
  - Ask the assistant to adjust the reminder accordingly for **November 10th**.
- **Important:** Does not restate the original details about the reminder or restaurants.

Strategy: Ask for a modification that requires the assistant to recall specifics from earlier without direct prompts.

Turn 10 (Assistant):

- Expected to confirm the updated reminder:
  - Acknowledging the combination of experiences from "La Petite Table" and "Le Nouveau Bistro".
  - Ensuring the reminder is still set for November 10th.
- **Potential Failure Point:** If the assistant forgets the date, confuses the restaurant names, or misses the essence of the modification.

Strategies and Tactics Summary:

- **Subtle Introduction:** Introduce key details early in the conversation naturally.
- **Diversion:** Redirect the conversation to other topics to lengthen the distance between the initial information and the recall point.
- **Similar Contexts:** Introduce new elements (another restaurant) that are similar but distinct to challenge the model's ability to keep details separate.
- **Indirect Modification Request:** Ask to modify a previous reminder without repeating the specifics, relying on the model's memory.
- **Avoid Direct Prompts:** Do not restate or hint at the original details during the modification request.

Target Endpoint (Failure Detection):

- **Success Criteria:** The assistant correctly modifies the reminder, incorporating both "La Petite Table" and "Le Nouveau Bistro" for November 10th, demonstrating accurate recollection of earlier details.
- **Failure Criteria:** If the assistant:
  - Confuses the restaurant names.
  - Forgets the original date.
  - Omits key details from the original reminder.
  - Provides a response that indicates a lapse in memory or context.

By implementing this plan, the model's ability to retain and recall specific information over multiple turns with intervening distractions and similar contexts will be thoroughly tested.

Figure 9: Initial Conversation Blueprint: Strategic Plan for Model Evaluation